Pratham Bansal

Systems Engineering & Design • Full-Stack Software & Hardware Developer
• Vernon Hills, IL • prathamb426@gmail.com • **https://pratham-bansal.com** •
https://www.linkedin.com/in/pratham-bansal-958174323 • https://github.com/pbansal426

Portfolio

EDUCATION

University of Illinois Urbana-Champaign — **Expected May 2030**
Bachelor of Science in Systems Engineering and Design — Urbana-Champaign, IL

EXPERIENCE

XPerience Consulting 501(c)(3) — **July 2023 – May 2026**
Intern — Vernon Hills, IL
- Created instructional coding content, mentored students, and helped organize educational events for local youth.

NectarSpot — **July 2025**
Marketing & SEO Intern — Remote / Vernon Hills, IL
- Completed structured SEO training covering keyword research, on-page optimization, and search ranking fundamentals — applying concepts to analyze and improve content discoverability for NectarSpot's platform.

Hack VH & D128 STEAM Showcase — **Oct 2023 – Mar 2025**
Web Developer — District 128, IL
- Architected and deployed official web platforms for a district-wide STEAM competition and 24-hour hackathon utilizing HTML, CSS, and JavaScript.
- Served as primary digital hub for event organizers, managing critical schedule announcements, resources, and registration workflows for 150+ participants.

KEY PROJECTS

MediCube — Smart Pill Dispenser & Support System — **Feb 2025**
Arduino, C++, Python, Computer Vision / AI — 2nd Place — Engineering (Hack VH)
- Built a physical pill dispenser using an Arduino with compartments, LED indicators, a speaker, joystick controls, and a text display.
- Connected the device to a web portal so family members and doctors can remotely set schedules, monitor usage, and get alerts — awarded 2nd Place in Engineering at Hack VH.

Streamline — AI-Assisted Audio Synthesis Plugin — **2024 – 2025**
C++, JUCE Audio Framework, Digital Signal Processing (DSP) — Desktop VST/AU Plugin
- Built a music production plugin in C++ and JUCE that runs inside professional audio software (like Ableton or Logic Pro) and uses AI to generate audio and MIDI patterns based on musical inputs — compatible with both Mac and Windows.
- Handled all low-level audio processing to keep the plugin responsive in real time, ensuring it works reliably inside a live recording session without causing lag or glitches.

Flowlyst & Lock In — Student Productivity Applications — **2023 – 2025**
Python, Flask, JavaScript, HTML5/CSS3 — Congressional App Challenge Entry
- Architected two independent full-stack web platforms using Python/Flask backends and vanilla JavaScript frontends, implementing task management, Pomodoro timers, focus-mode website blocking, and peer accountability systems from scratch.
- Submitted Flowlyst to the 2023 Congressional App Challenge (nationally competitive federal program); showcased Lock In at the 2025 D128 STEAM Showcase to 200+ community members and industry judges.

Caliper — RAG Diagnostic Intelligence — **Jun 2026 – Present**
Python, LangChain, ChromaDB, Sentence Transformers — Independent Project
- Building a diagnostic & repair service for car owners that ingests official documents from NHTSA into a vector database for semantic retrieval.
- Database connects to a custom-made RAG pipeline and evaluation system to ensure deterministic, high-accuracy diagnosis and repair guidance.

macOS AI Assistant Daemon — **Aug 2026 – Present**
Python, Ollama, macOS Accessibility API, SQLite — Independent Project
- A background agent that intelligently utilizes local language models to control computer with sub-second latency. Accessible to chat via hotkey, similar to Spotlight Search on MacOS.
- Developed a registry of 35+ system tools, including web search, file manipulation, and system control. Protected by a tiered permission gate to achieve correct tool-calling accuracy in custom benchmarks.

TECHNICAL SKILLS

Languages: Python, C++, C#, JavaScript, TypeScript, HTML5, CSS3, SQL
Frameworks & Libraries: React, Flask, Node.js, Express, Tailwind CSS, REST APIs, Electron
Hardware & Audio / Media: Arduino, Embedded C++, JUCE Framework (Audio DSP & Plugins), Unity Engine
Developer Tools: Git, GitHub, VS Code, Linux/macOS CLI, Docker, SQLite
AI Coding Tools: Claude Code, Antigravity